

Software Tools (BINF*6210) – Assignment #4 (35%)

DUE DATE: Friday December 5, 2025 11:59 PM to CourseLink Dropbox

Overview: Over the course of completing Assignment #4, you will conduct a small project on a topic that is useful or meaningful for you. You will either set your own question or objective, and you will adapt and build upon skills and concepts learned in class to solve new problems. Completion of this project involves synthesizing concepts and adapting skills you have learned throughout the semester.

Please see file #1 for candidate project ideas. Where relevant, part of your assignment may involve adapting class example scripts (or concepts), but it is also important to have a component of building onto that foundation by using a new package, analysis method, or visualization type, especially if you will use It is also important to implement good programming skills (such as creating functions and avoiding repetitive code). You are also welcome to propose your own topic, but you must speak with me for project approval at least three weeks before the assignment due date. Before making your project selection, I encourage you to read over all of the project ideas carefully, and also check out the provided links. It is useful to be aware of this array of resources.

To ensure your success and minimize stress, start on this project right away! This project is meant to be completed over 3-4 weeks of part-time but regular effort. Assignment #4 serves as both a learning opportunity and an assessment tool. This assignment also provides training, serving as a valuable stepping stone towards completing larger projects (e.g. BINF*6999 or your thesis and beyond).

Completing this project contributes towards achieving all five of our course-level learning outcomes:

1. obtain data from key databases relevant for bioinformatics and to understand the sources and limitations of these data

2. filter, manipulate, analyze, and visualize bioinformatics data
3. conduct reproducible analyses and use software tools for version control and collaboration
4. understand and apply selected algorithms commonly used in bioinformatics, including for sequence alignment and clustering
5. adapt the above skills to learn new tools and conduct new analyses not explicitly covered in class

General Guidelines

For most class members, your project should be written entirely in R. It is required that either all or the majority of your assignment be written in R. If suitable for your project (e.g. see file 1 with candidate project topics), you may use a software tool outside of R as part of your work. It is essential that you explain clearly what you did in the other tool. Also, you must provide the output from that tool, as an input for your R script.

Requirements and Project Steps for your story board:

1. Title slide: Title, name

2. Introduction (3 or 4 slides): Pose a question or objective that interests you, and set up your project. Using one of the themes outlined in file 1, or your own project idea (pending approval), narrow into a specific project objective and data set. Your project may involve building a tool (e.g. classifier), exploring an idea using visualizations, testing a specific biological hypothesis by applying analytical methods, or exploring the impact of analytical choices upon results. Be clear in your introduction: Which of these types of projects are you conducting? And, what is the objective of your project? Aim to cite 2-4 suitable references in your introduction. You may use any of the references uploaded to CourseLink or other literature you find on your topic.

Tips: The following is one possible way to organize your introduction into a nice flow.

Slide 1: What is the overarching topic you are interested in? Why is this scientifically important and interesting and/or societally relevant?

Slide 2: You can outline what is an important sub-area of research or a gap in knowledge. (As an example, perhaps your broader theme is gene expression analysis, and in paragraph 2 you could narrow in to introduce the importance of statistical choices.)

Slide 3: What is the specific objective of your study? What kind of study are you performing (e.g. Exploratory? Software comparison? Hypothesis testing? Classifier creation? Learning about tools to re-create a published figure?) What will you do?

3. Description of Data Set (1 or 2 slides). Provide a short written description of your data set.

What data set have you chosen to analyze to address your question? Describe the data set.

Where are the data from, e.g. from the literature? from a public database? On what date did you obtain the data? How did you obtain the data? What are the properties of the data set?

How many samples are there, and what are the key variables you are interested in? If you are comparing groups (e.g. treatments, organisms, cell types, genes), describe the nature of the groups that you are comparing. Cite the source of the data set.

I expect that most students will analyze a real, publicly available biological data set for this assignment. If suitable for your project, it is permissible to simulate data for your study, in addition to using a biological data set. If you wish to consider data simulation, I suggest speaking with me in advance. Also, if suitable for your project, it is permissible to use a dataset provided by your advisor, to gain experience analyzing a type of data you will be working with in the future. (However, note you can't copy/paste among courses or to your thesis, so speak with me if you have any doubt about what is acceptable.) If so, you need to explain the source of the data. Depending upon the size of the data set, you may consider analyzing only a subset here, to keep the project scope manageable. Data should be analyzable on a desktop computer for this assignment. As well, please note that you will need to provide a data file to enable your code to run. If unpublished data are included with your assignment, I would treat such data as

confidential, but you can ask your advisor to speak with me if they have any concerns. Please be sure to mention if you are using unpublished data in your assignment to ensure I am aware. (Students often bring to my attention interesting software tools or public vignettes that I may wish to check out, so I would want to know if there is anything confidential about your project.)

4. Write code to answer your question in a .R or .rmd file. Follow the “Best practices in R” resource on how to structure, format, and comment your code. If your project involves testing a hypothesis, then your code should involve a statistical test (such as linear regression if you are testing for an association between two continuous quantitative variables). Test that your code works as intended and is also well formatted before submitting!

You **should use GitHub** to manage your own work. You may keep your repository private. Include your GitHub repository link at the top of your assignment.

You may heavily draw from a Bioconductor vignette for your project, if suitable. Of course, you must cite any such sources used. However, running an existing vignette, with the provided data, end to end would not be a sufficient project. You must build upon such tutorials and include a novel component for your project. Many examples are provided in file 1. Examples include: comparing the results from using two packages, testing a biological hypothesis using a different data set compared to what is used in the vignette, combining two software tools for a novel project (e.g. combining DADA2 output with a phylogenetic community analysis, such as using phyloseq or picante), trying a range of parameter settings for selected functions and exploring the impact of methodological choices upon the results, or re-creating all or part of an interesting analysis or figure from a publication using your own code.

Whenever possible, include code for data acquisition for your project. Also, include all data files needed for analysis with your submission.

After data acquisition, you will next conduct data exploration, filtering, and quality control. What is suitable in this section will depend upon your project. Here are a few examples:

- For key quantitative variables (e.g. organism length, mass, etc.), calculate data summaries, such as using the summary() function. Also, prepare a suitable plot, such as

a boxplot or histogram or violin plot, showing the distribution of the data. If suitable for your project, you may use multiple panels in a single figure if you have multiple variables of interest. Check on outliers. Are these likely real data points, or could they be data entry errors? Note that we are looking for errors, not looking to exclude data that disagree with a specific hypothesis. You may decide, for example, that even if there is an outlier that it is likely real data and you might perform a data transformation so that the observation doesn't have extreme influence upon your results. Or, you may perform your analysis with vs. without extreme data points for comparison.

- For DNA sequence data, you may consider doing a check for sequence lengths and exclude sequences having very short length compared to the majority of your data, for example. The suitable choice would depend upon what gene and gene region you are analyzing. For example, for COI data for animals, there is only a very small amount of natural length variability (and thus very short sequences would be due either to poor quality or to metabarcoding projects which may involve sequencing short gene fragments). By contrast, in markers such as 18S, there can be a large amount of natural sequence length variability to consider. You may also consider excluding sequences with a large proportion of Ns. A suitable visualization for DNA sequence data could be a histogram of sequence lengths. If suitable for your project, you may investigate outliers based upon a pairwise distance matrix built either from k-mer frequencies or aligned sequences (i.e. are there sequences very different from others in the dataset?). Another helpful quality control step could be to build a visualization (such as a dendrogram, including scale bar of branch lengths) to look for sequences very different from others. You could then BLAST unusual sequences (using NCBI tool) and check the top few hits and see if there could be a misidentification in the dataset (or another serious issue, such as the wrong gene, a different gene region being sequenced, wrong sequence orientation, wrong organism/contamination, etc.).
- For RNA-Seq data, you should examine library size among samples and make a suitable analytical choice that takes into account variability in library size. A suitable plot for that type of analysis could be to plot gene expression levels before vs. after normalization

(e.g. see the gene expression vignette we ran together in class). You should also explicitly consider how you will choose to treat lowly-expressed genes for your project.

The above are a few examples. In general, for your project, think (and read) about: What can go wrong with the type of data you are analyzing? And, what types of variability might be expected in that type of data?

Also, make a deliberate decision about how you will treat missing data for your project. Will you focus on genes or traits with a good sample size, for example? Do you need complete cases (e.g. can you only include species having sequence data for each of two genes)? If suitable for your project, you may consider incorporating imputation techniques.

You would then move on to performing your main analysis. What would go in this section would depend upon your project, e.g.: writing code to build and test a classifier, build visualizations to explore ideas, conduct a statistical test of a biological hypothesis, or explore the impact of methodological choices upon results.

5. Methods slide. Visualize your pipeline.

6. Figures. Your storyboard must include **4-6 figures**, one on each slide, with as header what information the audience should extract from the figure. This can be a combination of exploratory and results figures, but with a focus on results: your project should be extensive enough to result in several important figures that answer your biological question. Throughout, ensure that your figures are clear and well labeled. Even for simple figures, such as histograms, ensure that you have accurate, informative axis labels. Include units of measurement in your axis labels. Also, consider readability, visual appeal, and accessibility. Use well-differentiated colours, and avoid relying upon the red-green spectrum to convey scientifically important information. Remember, you can consider using a combination of colour and symbol/pattern to convey your meaning. The grade in this section is based upon quality and novelty, not having the maximum permissible number of figures.

7. Results and Discussion (4 or 5 slides). At the end, write a short section describing and interpreting what you discovered through conducting your project.

Please find here an example outline that would provide a nice flow for this section:

- Slide 1-2: Return to your original question. What is the answer to your question? What did you discover? Were your results as expected or not?
- Slide 3: Briefly describe any key caveats of your study. For example, are the conclusions that can be drawn limited by sample size or any other concerns? Were there biases in data availability that could have impacted your project?
- Slide 4: What would be the next steps for this research? What would you do next if you had more time and if you were going to develop this work into a larger project? Did your results reveal any interesting preliminary findings that would be worthy of follow-up study?

Tip: I suggest that you consider citing 2-4 references in your discussion section to help with interpretation of your results. You may cite references that you also cited in your introduction.

8. Reflection (1 Slide) (3% bonus). Reflecting upon our learning process and progress is a valuable strategy, helping us to improve our ability to learn and to set our next learning goals. At the end, I encourage you to add a paragraph reflecting upon the process of completing this assignment. What did you learn through completing your small project and this course? What lessons will you take forward in your future coursework, BINF*6999 or thesis, and future career? What skills do you want to work on next to help you to meet your graduate program and career goals? You may consider reflecting upon your progress with technical skills as well as personal strategies, time management, etc.

9. Acknowledgements slide. If you received project tips from others or help solving problems, include that information in this section. We will have a class activity to discuss your project ideas; so, if you receive advice that helped you to develop your project, indicate so in this section. Briefly, indicate who you talked to, the nature of the advice, and how this impacted your project. You may speak to other class members. It is NOT permitted to complete the assignment for someone else or to copy blocks of code from others. If someone helped you to get unstuck when you were facing an error message, then indicate who and what you learned from this. (To clarify: You ARE allowed to talk to others, but you are NOT permitted to let them

fix the problem for you without you actually understanding what is going on. Write about: What was causing the error message, and what did you learn during the process of obtaining help to fix it?)

10. Reference slide. If you cite any sources from the scientific literature, cite them here. You may wish to consult 2-3 papers from the literature to help you to develop your idea and/or interpret your results. You would include such references as an in-text citation in the relevant sentence of your assignment, i.e. introduction or discussion (e.g., Xu et al. 2020). Also, list the full reference here at the end. Be consistent in your formatting. If you used any specific online tutorials or a specific StackOverflow posting, for example, also include those here.

Overall Scope and Length: Your final storyboard is expected to be approximately 18-20 slides in length. Convert your slides to a pdf file, and save it in the “doc” folder of your project folder structure. Your folder structure for this assignment should at least have a R, doc, and figs folder, but you can include the other suggested folders if your project requires this. Submit your folder structure as a zipped file to CourseLink, to ensure that everything is contained in one file that has all the components to run, evaluate, and interpret your assignment.

Tips for Success - Checklist

- ✓ Start your project early to avoid stress at the end.
- ✓ Before you start coding, sketch out the steps in your analysis. This helps you to think about the suitable ordering of steps. Also, this creates a checklist for you of coding tasks, each of which is more manageable. “Do Software Tools Final Project” is too large an item to put on your daily task list and makes it hard to get started. “Download cytb data for genus Rana from NCBI nucleotide database and check sample size” is a much more achievable task! Break your project down into small, achievable steps.
- ✓ See File #1 for a suggested timeline for working on your project.

- ✓ If you get stuck, first try to resolve the issue yourself: read the error messages carefully, google your error message, read the function documentation, see online tutorials, etc. Then, you may reach out to others for help to get past your issue. I encourage you to participate in the review/help sessions. You may also post questions to the Discussion board on CourseLink. Please note that if you would like to request help from either the instructors or your peers, please do so in advance, not the night before. Please proofread your assignment before submission. I recommend submitting early to avoid technical difficulties at the last minute.
- ✓ A storyboard is slightly different from a presentation. It uses the same format as a presentation (i.e., your choice of presentation software), but you have the flexibility of using more traditional bullet points to explain your decisions.
- ✓ Check that all components of your assignment match. For example, parameter settings mentioned in the commenting should match the implemented parameter settings in the code. The figure axis labels and figure title should match what the code is doing.
- ✓ Before submission, start a new RStudio session. Then, select the entire script (e.g. you can use the 'Control-A' keyboard shortcut) and run the entire script in one go. It should run without error messages. If you get an error message, fix that issue and then repeat until the full script runs.

Submission Instructions:

Submit your assignment as one zipped file to the designated Dropbox folder on CourseLink.

Software Tools (BINF*6210) – Assignment #4 Grading Rubric

This grading rubric is intended to help you to be as successful as possible on Assignment #5.

Instructor judgement will also be applied during grading. For example, high-quality or novel project components not explicitly mentioned below would also be considered during grading.

Note that each level is cumulative. That means you should meet the criteria of the lower level to move on to receiving a grade at the next level. Aim to complete all components at the first level (70%) first. Then, after you check off everything in the 70% category, work on your assignment further to achieve a higher grade by improving the quality, scope, novelty, or creativity of your project. This table is populated with examples of components that would earn a particular grade.

I have started the rubric at 70%. While 65% is the passing grade for an individual course at the graduate level at the University of Guelph, I assume you also want to do well overall, pass your graduate degree (at least 70% overall GPA required), and achieve a level of competency that will help you in your further studies and future career. You can do it! Your grade may fall in between two levels. This rubric is intended as a guide only to help you to be successful.

At the end of completing this assignment and course, I hope that you will be proud of your achievement. I suggest that you consider posting your work to your own GitHub page. Throughout your coursework and thesis/project work, I suggest that you work towards building a portfolio of scripts and projects to share with future advisors and prospective employers.

I hope that you enjoy completing this assignment! I look forward to reading your projects.

Assignment Component	Good (Grade of 70%)	Very Good (Grade of 80%)	Excellent to Outstanding (Grade of 90-100%)
Storyboard (/10)	7/10 *all the components of the storyboard are included *a clearly phrased question is posed *a statement is made about why this question is interesting *reader can follow what the project is about and what is the objective *writing is understandable, with few grammatical errors	8/10 *novel question is posed (may use example question from instructions, but project would go beyond what is covered in example scripts) *reference is made to the literature (e.g., 1-2 relevant references cited)	9-10/10 *novel question is posed *reference is made to the literature (e.g., 1-3 highly relevant references cited) *critical thinking displayed (e.g.,: outlining expectations under two contrasting hypotheses; that's just one example)
Description of data set (/5)	3.5/5 *basic information is provided about data set (e.g. data source, type of data)	4/5 *key variables of interest and data set attributes (e.g. sample sizes, groups) are clearly described	4.5-5/5 *thorough account of data set is provided
Code – Part 1 (Data Exploration) /25	17.5/25 *code is contained in one zipped file with all components * you followed all relevant best practices (see list below) *code must work as intended *code must run end-to-end without errors *data acquisition clear *summarizes key variables of interest (e.g. mean, range of	20/25 *clear explanations are included about data exploration and reasons for specific data filtering steps *includes thoughtful, deliberate approach as to how you will treat NAs in your variables of interest for analysis *thoughtful explanations are included about why you are doing what you are doing; considers settings for arguments used (if you use default,	22.5-25/25 *strong critical thinking demonstrated to explore potential errors or biases present in the source data that may influence your project *may include novel types of data summaries or graphics to show the distribution of data and to reveal potential errors or biases *will include some novel code

	numerical variables, sequence lengths) *check for evidence of serious errors in any variables you are analyzing *most lines of code may be adapted (correctly) from example scripts *use commenting to explain what you are doing at each step *code well formatted *code available on GitHub	that's fine, but explain why) *may include some code that is newly written (more than minor adaptation of provided example scripts) *low redundancy of code (e.g. instead, use apply functions or a loop or a vectorized mathematical function, depending upon what you are doing)	*low redundancy of code; may involve writing of own function(s)
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Code – Part 2 (Analysis to Address Question) /30	21/30 *code must work as intended *code must run end-to-end without errors *includes an analysis to address your main objective *most lines of code may be adapted (correctly) from example scripts	24/30 *includes an original component beyond example scripts, such as using a new method, software tool, or graphical approach *includes some lines of code that are newly written (more than minor adaptation of provided example scripts) *code has low redundancy (e.g. uses apply functions or loops or vectorized mathematical functions, as suitable for problem that needs solving)	27-30/30 *includes a component with strong novelty or creativity *uses one or more packages or functions or statistical methods not included in example scripts *may include a sophisticated graphical or statistical approach *code has low redundancy and high novelty (e.g. could include writing own function) *excellent commenting
Quality of Figures /20	14/20 *storyboard includes 4-6 figures *figures may be simple (e.g., histogram, boxplot, barplot), but should	16/20 *includes at least one type of either a novel figure compared to class scripts or sophisticated figure *colour, symbol shape, and/or pattern may be used in one or more	18-20/20 *all figures are polished, visually appealing, and clearly communicate the data and results

	be suitable for type of data being presented *axes should be clearly and correctly labeled *symbol size/style should be easy to understand	figures, with clear meaning	*creativity or novelty expressed in one or more figures
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Results and Discussion /10	7/10 *clearly state what you found, answer your original question *state whether your results were aligned with your expectations or not, why might that be?	8/10 *comment thoughtfully on any caveats of your study (e.g., limited sample size, sampling bias, etc.) *comment on what you would do next if you were to expand this study *references 1-2 relevant papers	9-10/10 *excellent critical interpretation of your findings and caveats *outlines next steps for this research (e.g. did the findings generate a hypothesis that could be tested in the future?) *refers to the literature to aid interpretation and discussion of next steps (e.g. 2-4 highly relevant references cited) *excellent quality writing, flows well, no or very few errors
Reflection /3 bonus	3/3 * reflect on what you learned with this assignment * what you will be using moving forward		

*The acknowledgements section is expected to be present for academic integrity, if you talked to anyone else about your assignment.

*Reference list must also be included. You should cite your main software tools used, any scientific literature cited (3-6 papers recommended, 10 maximum), any vignettes that you consulted, online tutorials, specific StackOverflow postings, etc. Grade credit for the quality and relevance of the reference list is included in the Introduction and Discussion Grades.

Total Graded out of 100% and Valued at 35% of Course Grade

Best Practices	Complete?
Visualize pipeline	
Ensure all steps are included and commented appropriately	
Include all information on dataset, packages used, and choices made	
Cite all packages used and any other pipelines consulted	
Consistent code formatting	
If necessary, start with a small dataset to test code before scaling up	
Include steps to explore and filter dataset	
If necessary, include steps to optimize parameters or do sensitivity analysis	
Create figures as checks within the code	
Include performance checks to ensure code is working properly (e.g., check if NAs are removed after the code)	
Include reproducibility checkpoints (e.g., sample size after filtering for a standard data set)	
Open pipeline up for review	
Keep track of changes and versions	
Make pipeline publicly available	
Ensure pipeline is general enough to be used on multiple datasets	
Make the pipeline as efficient as possible (loops, down-sampling, multiple cores, ...)	